1. Introduction & Requirements:

The objective of this project was to design and manufacture a lightweight multifunction wrench handle capable of interfacing with detachable crowfoot wrench heads while being manufactured using FDM additive manufacturing. The project was completed for the ME 313 Design of Mechanical Components course.

The customer requirement was to produce a lightweight and durable wrench handle suitable for applications where minimizing mass while maximizing functionality is critical. The handle needed to securely retain interchangeable crowfoot wrench heads without the use of adhesives or fasteners.

Group 33 was assigned the following wrench head sizes:

- 7/16 in crowfoot wrench head
- 1/2 in crowfoot wrench head

The final design was manufactured from ABS thermoplastic using a Stratasys F170/F190 FDM printer.

Project Requirements

The final design was required to satisfy the following technical requirements:

- Total handle mass less than 70g
- Manufactured using ABS on a Stratasys F170/F190 printer
- Must fit within the printer build volume of 150 mm × 40 mm × 20 mm
- Must interface with McMaster-Carr detachable crowfoot wrench heads
- Must retain wrench heads without glue or mechanical fasteners
- Must include at least one additional tool functionality
- Must have a flat side for stable placement on a table
- Must withstand expected torsional and bending loads during operation
- Must have no sharp edges
- Must be suitable for additive manufacturing

Additional Team Design Requirements

In addition to the project requirements, the team established several internal design objectives:

- Minimize stress concentrations using fillets and smooth geometry transitions
- Reduce unnecessary material to decrease overall mass
- Maintain adequate wall thickness near the wrench head interface
- Ensure all features are printable without excessive support material
- Design for repeated wrench head insertion and removal

2. Concept Generation:

Initial Concept 1 – Andrew_Wrench_V1

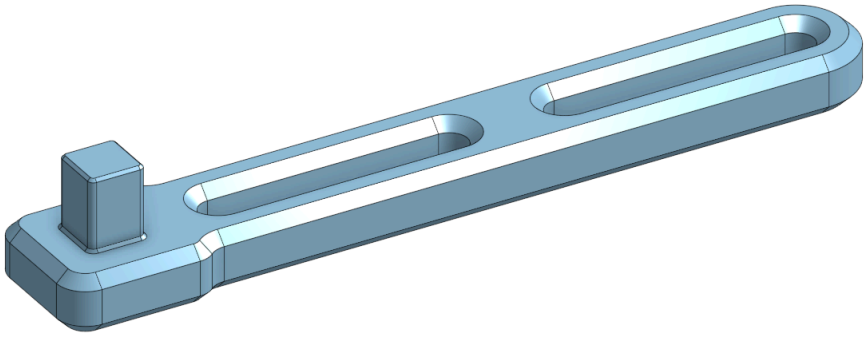


Figure X: Andrew's initial wrench design

The first design created by the first group member, Andrew, focused primarily on developing a lightweight wrench body while maintaining sufficient structural stiffness for basic loading conditions. The design utilized a simple rectangular geometry with a symmetrical handle profile to simplify both modeling and additive manufacturing. A press-fit interface was incorporated at the front of the handle to retain the interchangeable crowfoot wrench head, with a small fillet added at the base of the wrench head pin transition to help reduce stress concentrations in the region expected to experience the highest loading. The overall geometry was intentionally kept simple and avoided large overhangs so the part could be printed easily on the MIE machine shop printers with minimal support material and reduced print complexity.

To minimize mass while not impacting the max stress significantly, two large cutaway regions were incorporated into the center section of the handle. These cutaways significantly reduced the overall weight of the component while attempting to avoid major increases in maximum stress within the structure. The internal cavities created by these cutaways also served as integrated storage locations for screwdriver bits, satisfying the required secondary functionality of the project. This initial design iteration strongly prioritized weight reduction and manufacturability, resulting in a total part mass of only 24.17 g compared to the project limit of 70 g.

Initial Concept 2 - Olga-Dimitra Stoura_Wrench

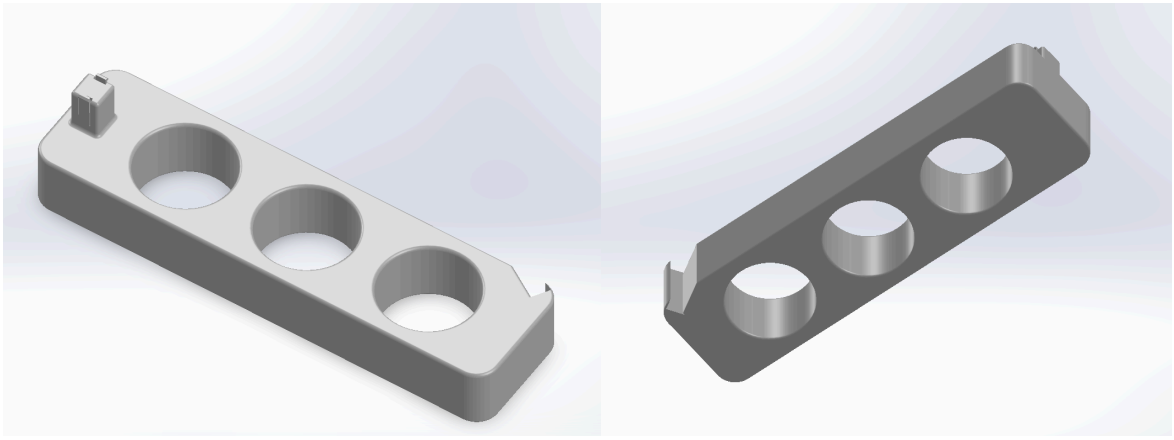
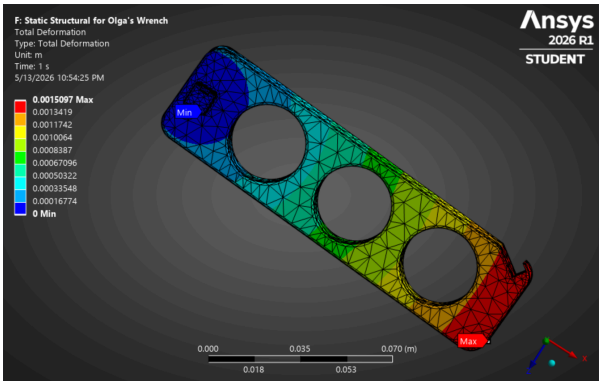


Figure X: Olga's initial wrench design

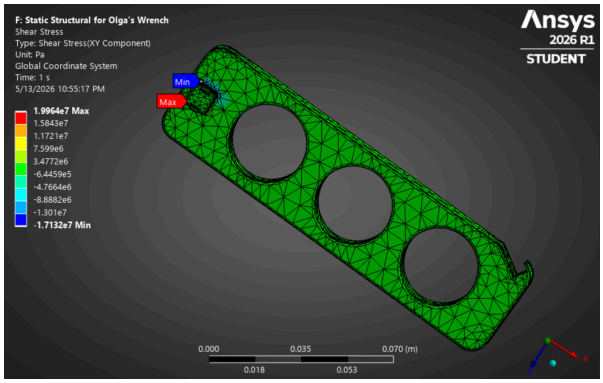
The second initial concept, created by Olga, focused on structural rigidity and functional integration while still maintaining a relatively lightweight geometry that is suitable for manufacturing. The design adopted a thicker rectangular body with rounded outer edges to improve overall durability and reduce localized stress concentrations along the handle.This design used a more solid cross-section with fewer aggressive cutaways, prioritizing stiffness and load distribution throughout the wrench body. A press-fit feature for the interchangeable crowfoot wrench head was incorporated at the front corner of the design, while the opposite end of the handle integrated a bottle opener feature to satisfy the project's secondary functionality requirement. The overall geometry was intentionally kept simple to ensure reliable manufacturability on the MIE machine shop 3D printers while minimizing the need for support material during printing.

To further reduce mass while maintaining structural integrity, three circles were cut throughout the wrench's body. The circular geometry of the cutouts helped promote smoother stress distribution through the handle compared to sharper internal features, reducing the likelihood of stress concentrations developing around the storage regions. This initial design iteration therefore

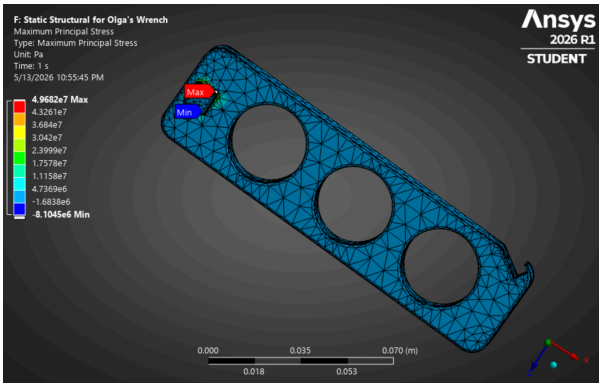
balanced manufacturability, stiffness, and multifunctionality, resulting in a robust and practical early-stage wrench configuration.



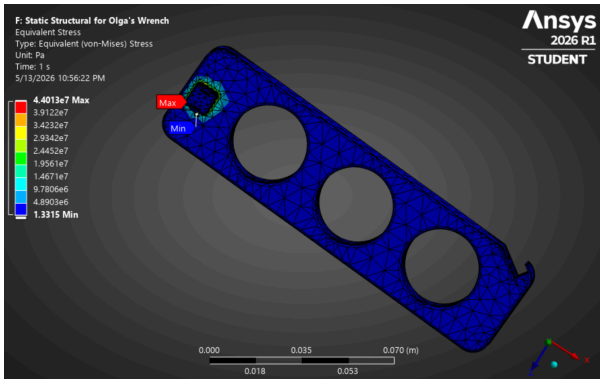
Total Deformation



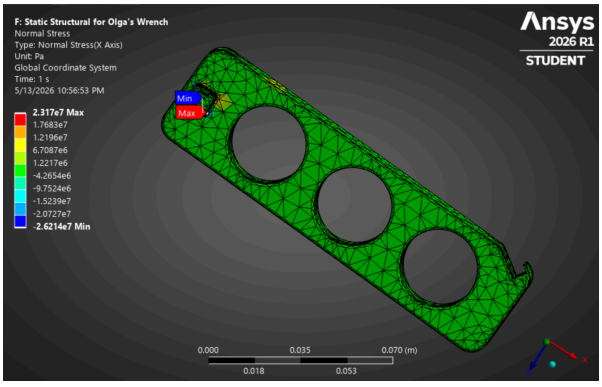
Shear Stress



Max Principal Stress



Equivalent Stress



Normal Stress

This analysis mimics the torque applied about the center drive of the wrench. This exploits the weakness throughout the wrench when turning it with a 25 N force. Based on these results, the stress is concentrated around the base of the drive, which is to be expected.

Figure X: Ansys analysis of Olga's initial design

Initial Concept 3 - Luke Goodwin_Wrench

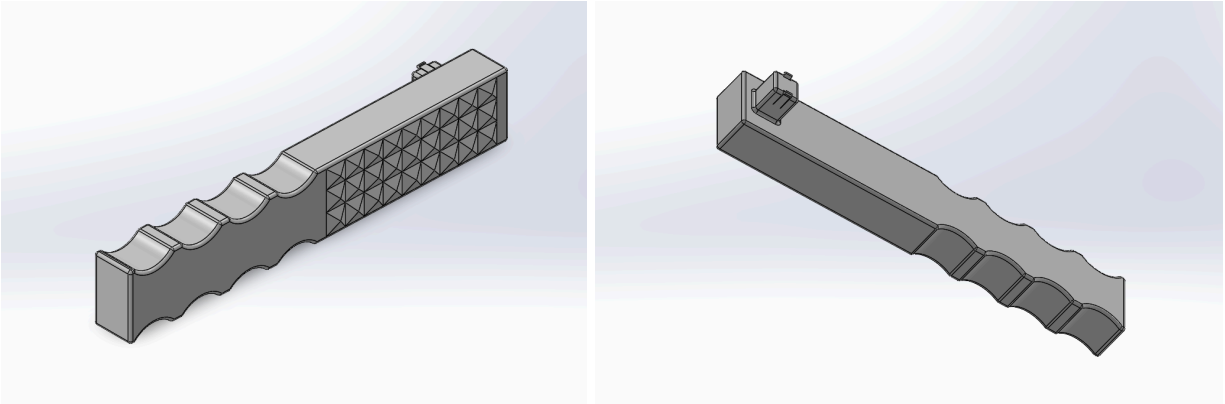
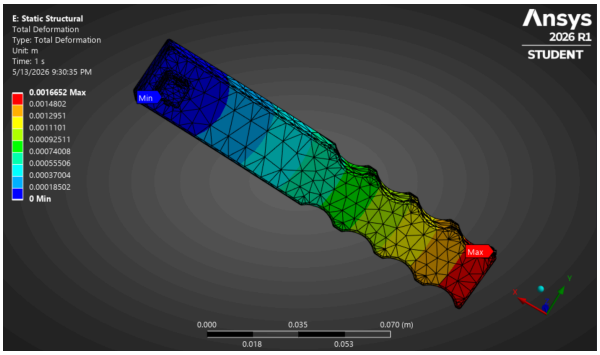
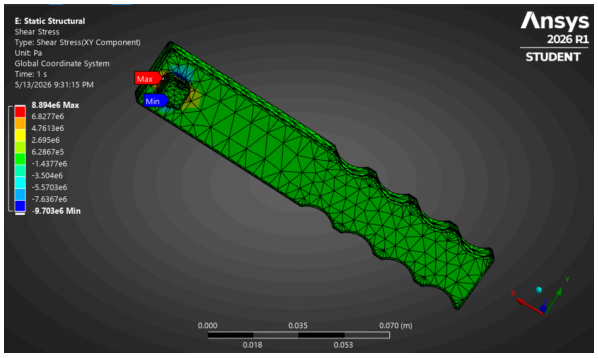


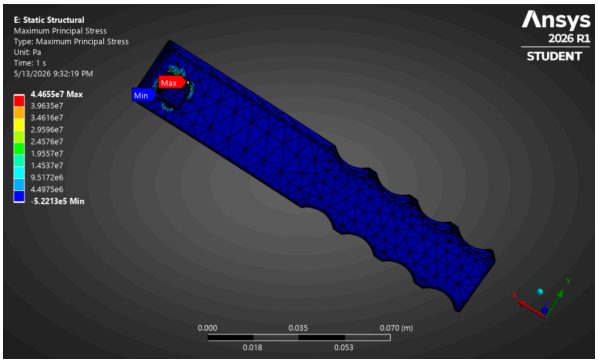
Figure X: Luke's initial wrench design



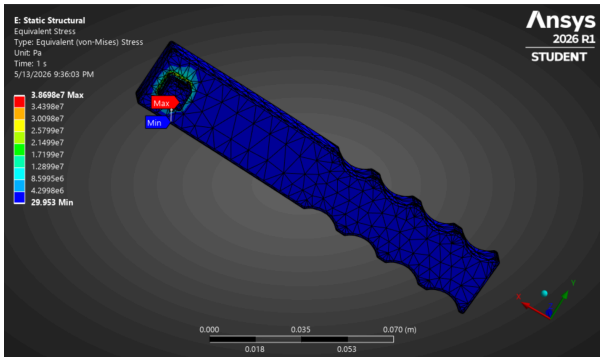
Total Deformation



Shear Stress



Max Principal Stress



Equivalent Stress

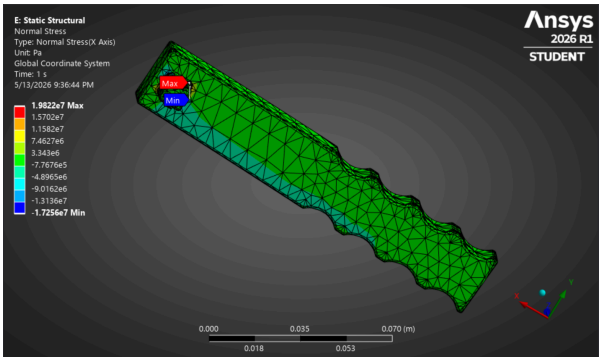
 Normal Stress	This analysis mimics the torque applied about the center drive of the wrench. This exploits the weakness throughout the wrench when turning it with a 25 N force. Based on these results, the stress is concentrated around the base of the drive, which is to be expected.

Figure X: Ansys analysis of Luke's initial design

3. By hand analysis set up:

Assumptions:

1. The handle behaves as a cantilever beam with a uniform cross sectional area.
2. The loading is static.
3. ABS material is homogeneous and isotropic.
4. Stress concentrations from layer lines are neglected.

5. The critical loading condition occurs at the wrench head interface.
6. Torque is transferred uniformly through the cross section.
7. The handle cross section is approximated as rectangular.

Material properties:

Ultimate tensile strength: 28.1MPa

Yield strength: 30.8MPa

Elastic Modulus: 2.4 GPa

Shear modulus: 0.85 MPa

4. Finite element analysis

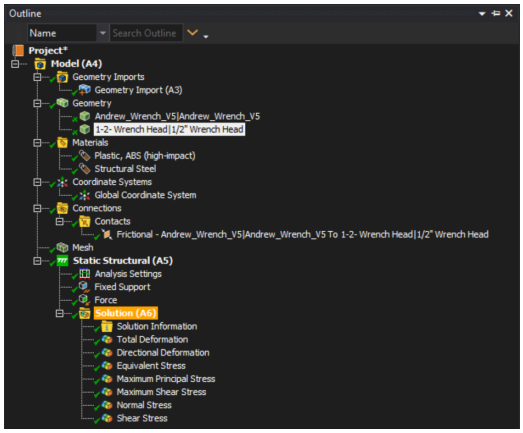
Assumptions Used:

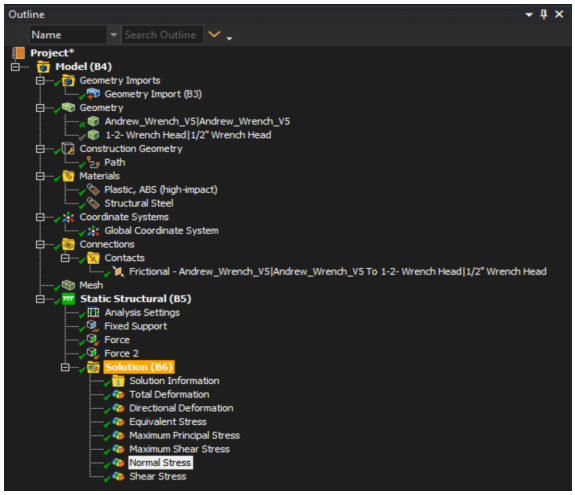
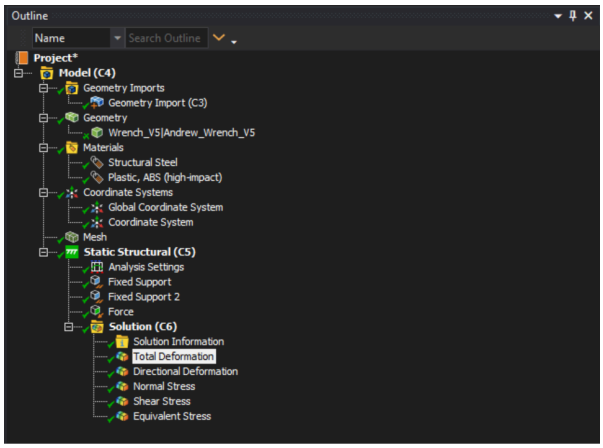
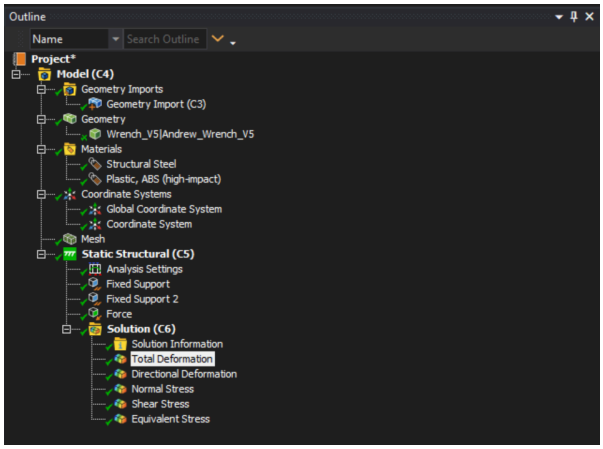
- For all tests(including Ansys for other designs)
  - Wrench Geometry is assumed to represent the exact dimensions of the manufactured wrench.
  - The provided ABS plastic in Ansys represents the manufacturing material(property values are similar).
  - ABS plastic is assumed to be homogenous and isotropic.
  - All applied forces are static and constant.
  - Deformation may be exaggerated based on mesh

- For Wrench handle + head tests
  - Only the wrench body is being studied, the wrench head will be hidden for all solutions.
  - Forces are applied at the very end edge of the wrench handle.
  - The wrench body and head are separate entities that use a friction force connection for slippage.
  - Structural Steel is an adequate material for the wrench head that is stronger than ABS.
  - Fixed support for the wrench body will be the entire wrench head itself.

- For Wrench handle alone
  - Fixed supports are located at the highest edge at the ends of the wrench.
  - All translational and rotational DOF are restricted at fixed features.
  - Force is a point force approximated at the center of the top face or side face.

Setup:

Wrench Handle + ½ inch wrench head with applied torque:	

Wrench Handle + ½ inch wrench head with applied torque and perpendicular load:	
Wrench Handle 3 point bending test in the z direction:	
Wrench Handle 3 point bending in x direction:	

Wrench Handle + ½ inch wrench head with applied torque:

1. Total Deformation

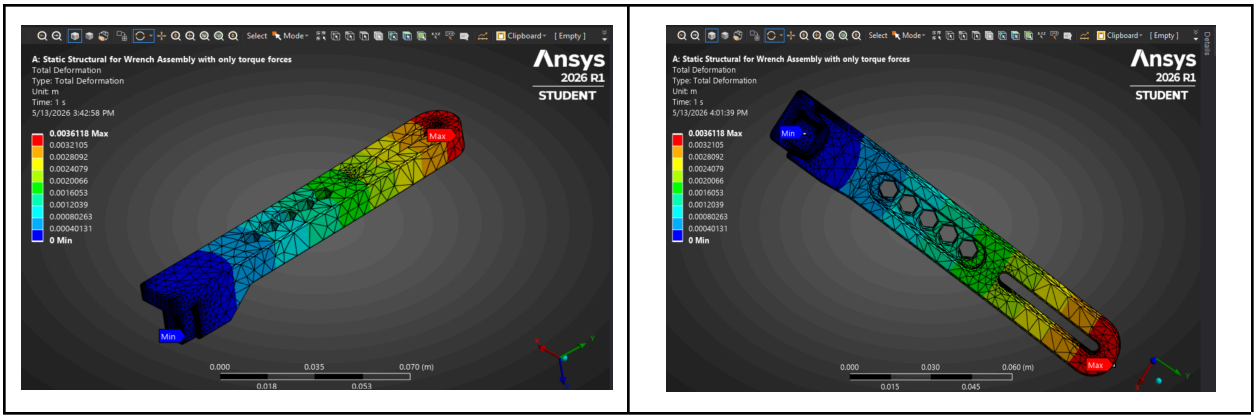


Figure X: Ansys total deformation with applied torque

2. Equivalent Stress

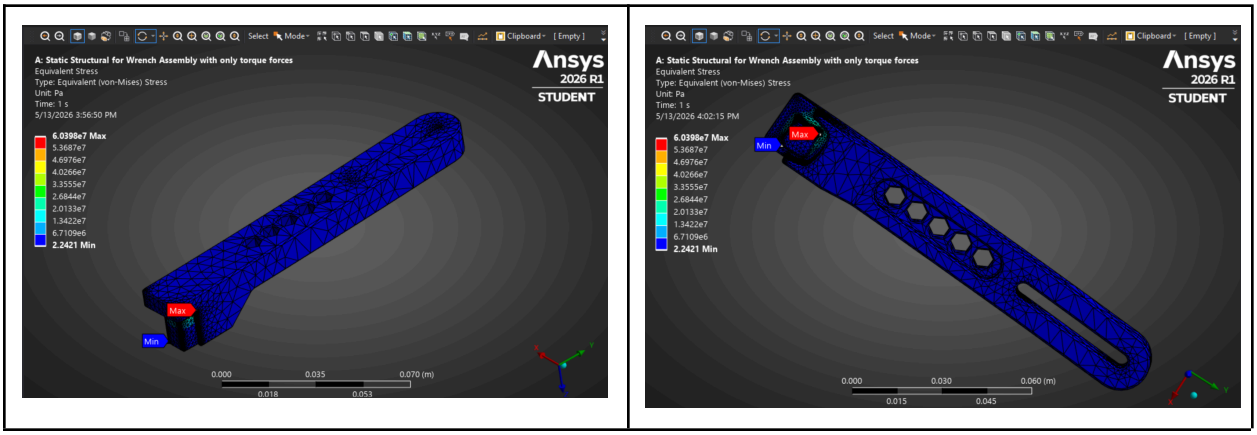


Figure X: Ansys equivalent stress with applied torque

3. Max Principal Stress

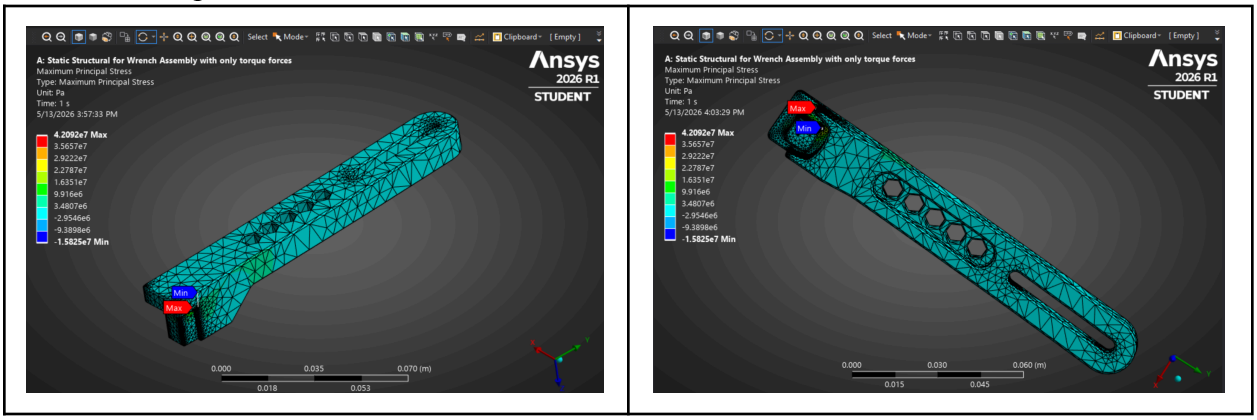


Figure X: Ansys max principal stress with applied torque

4. Max Shear Stress

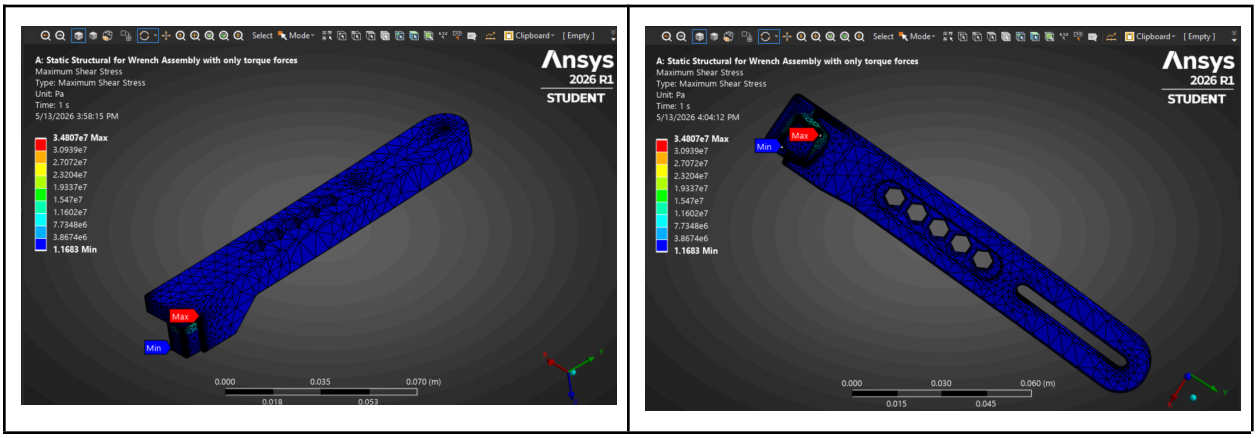


Figure X: Ansys max shear stress with applied torque

5. Normal Stress

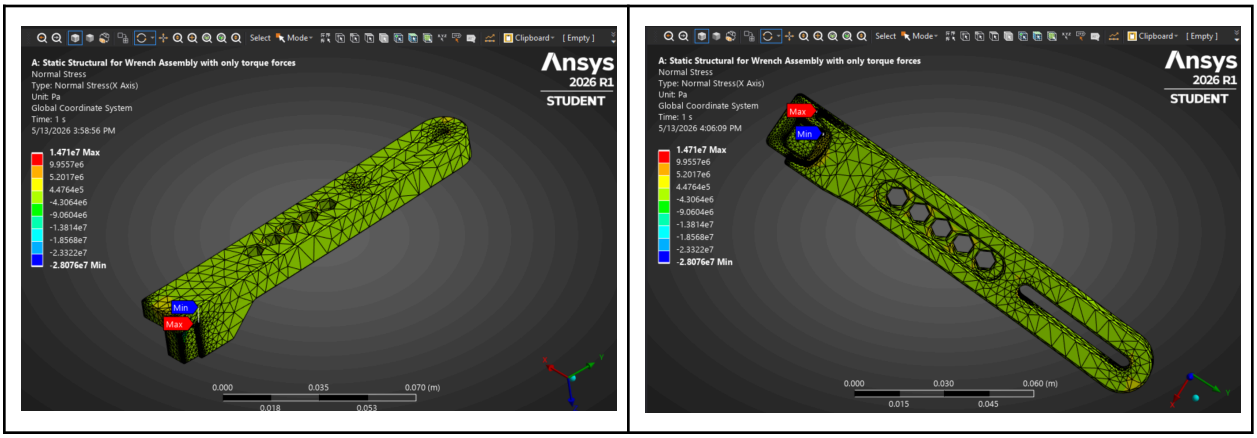


Figure X: Ansys normal stress with applied torque

6. Shear Stress

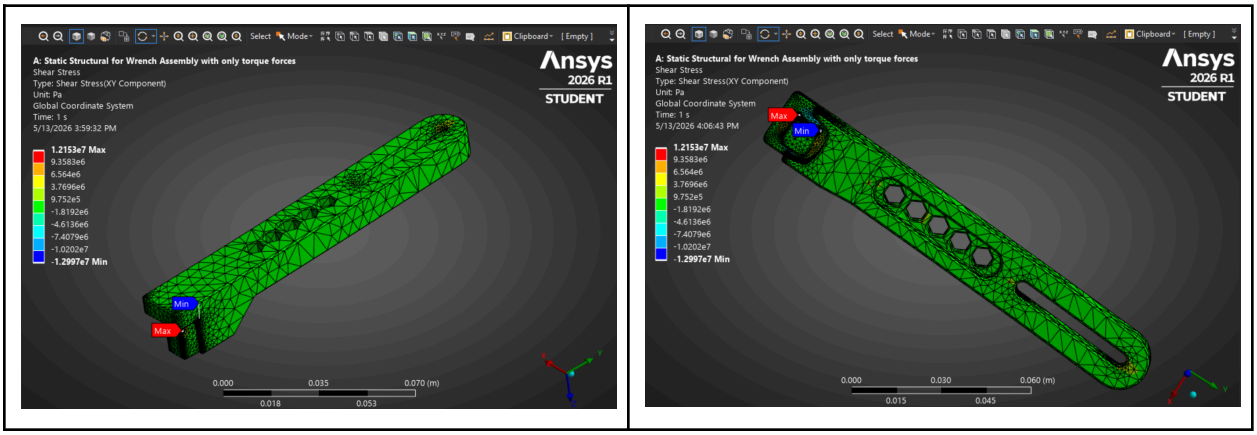


Figure X: Ansys shear stress with applied torque

Wrench Handle + ½ inch wrench head with applied torque and perpendicular load:

1. Total Deformation

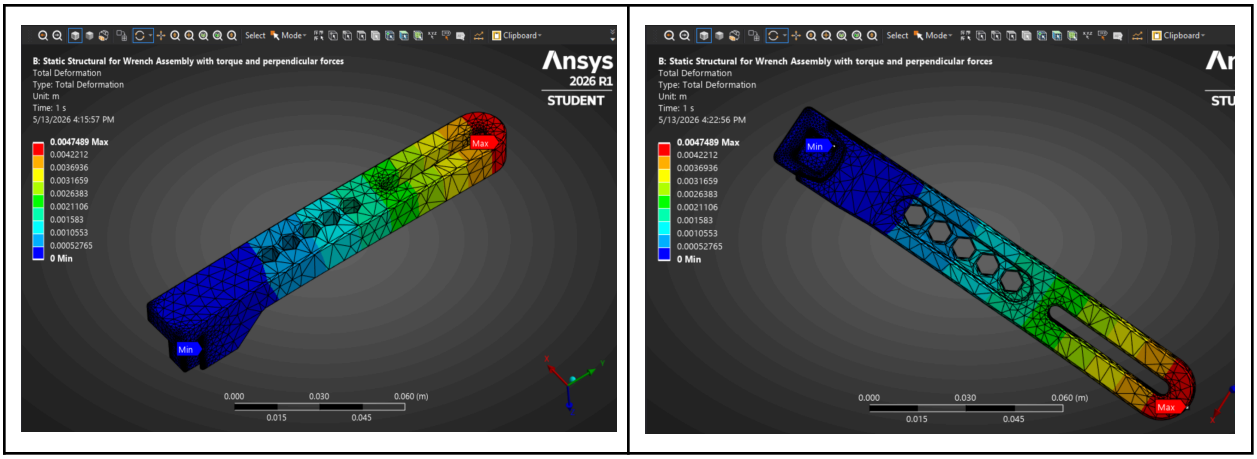


Figure X: Ansys total deformation with applied torque and perpendicular load

2. Equivalent Stress

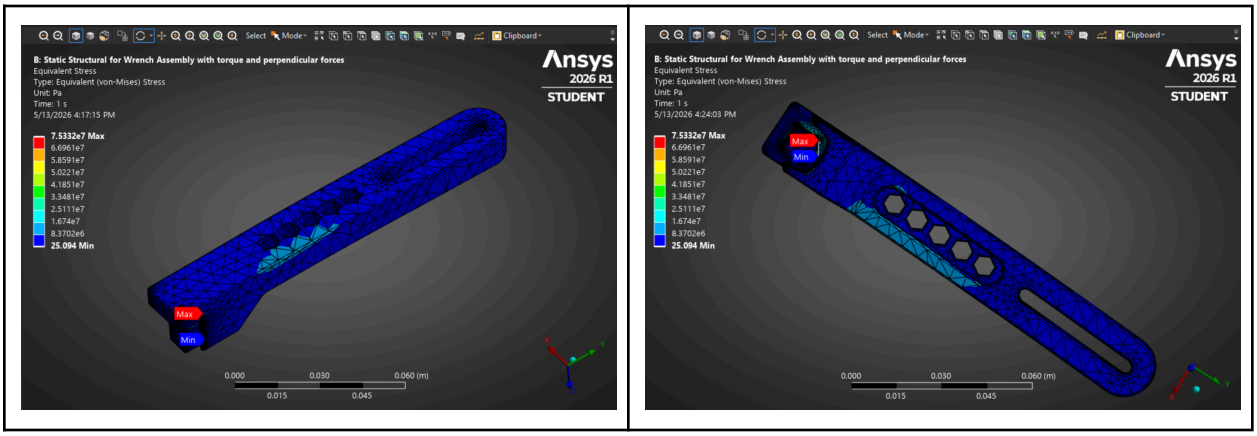


Figure X: Ansys equivalent stress with applied torque and perpendicular load

3. Max Principal Stress

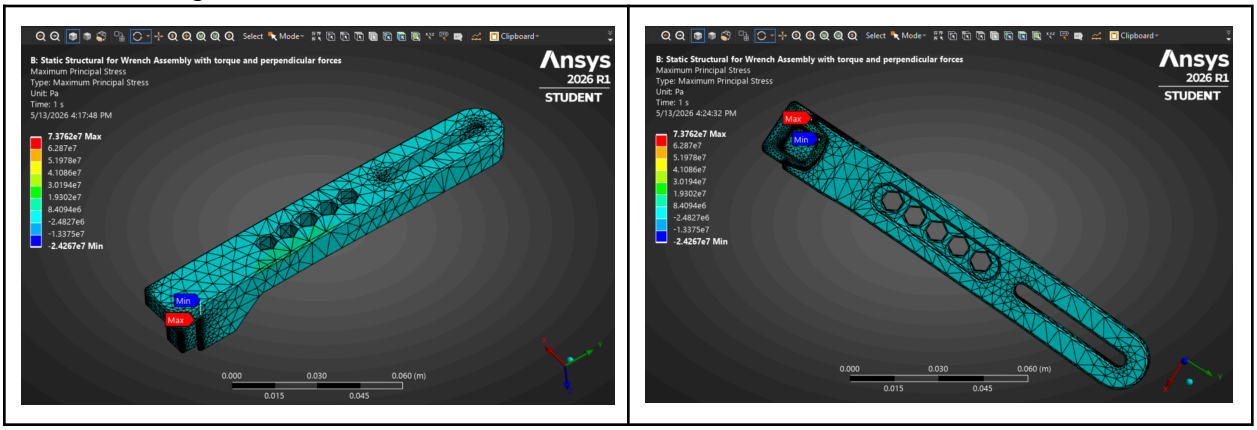


Figure X: Ansys max principal stress with applied torque and perpendicular load

4. Max Shear Stress

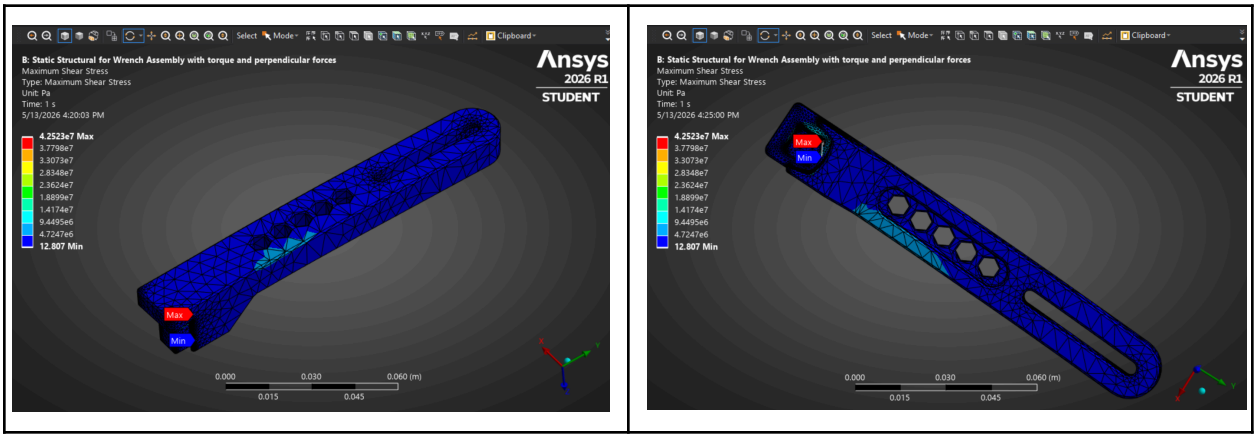


Figure X: Ansys max shear stress with applied torque and perpendicular load

5. Normal Stress

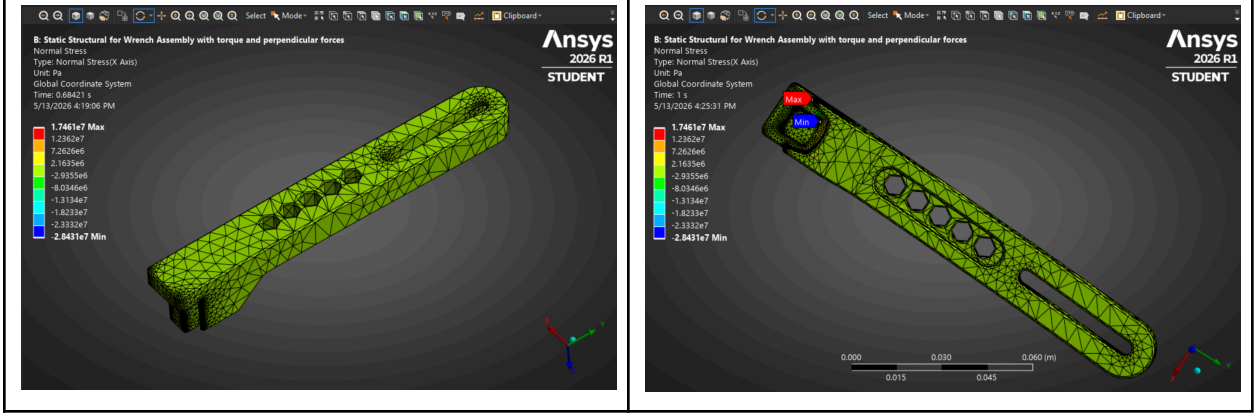


Figure X: Ansys normal stress with applied torque and perpendicular load

6. Shear Stress

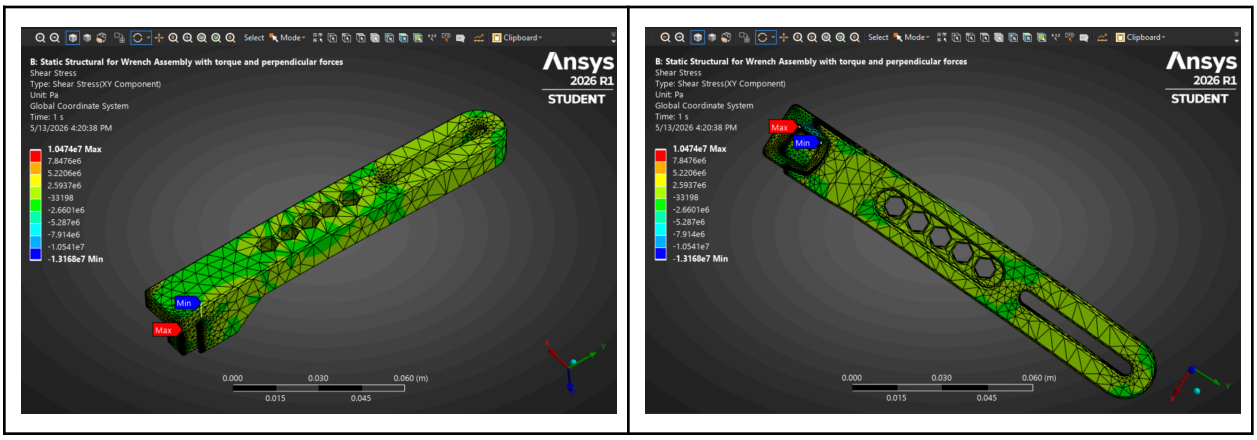


Figure X: Ansys shear stress with applied torque and perpendicular load

Wrench Handle 3 point bending test in the z direction:

1. Total Deformation

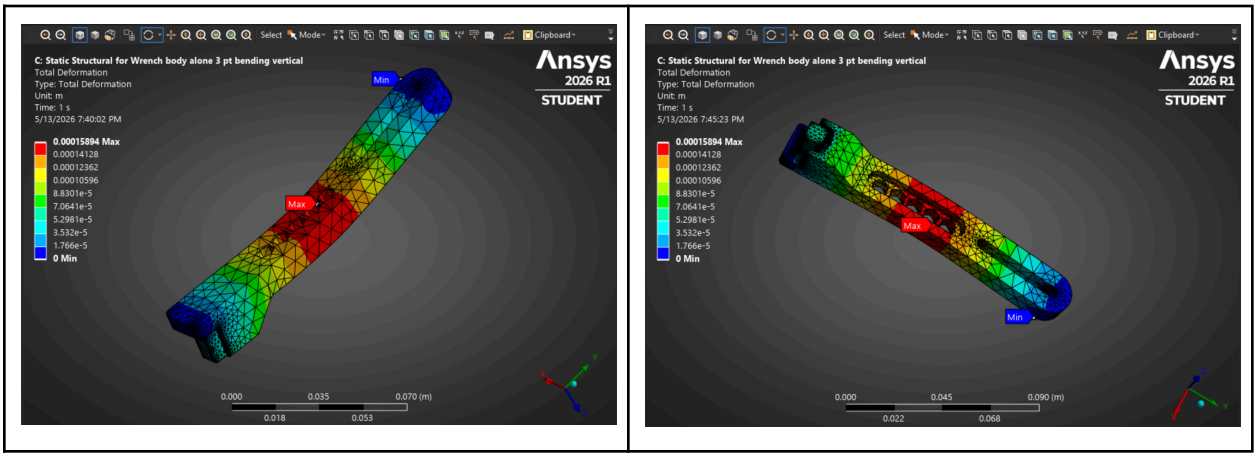


Figure X: Ansys total deformation with 3 point bending test in the z direction

2. Shear Stress

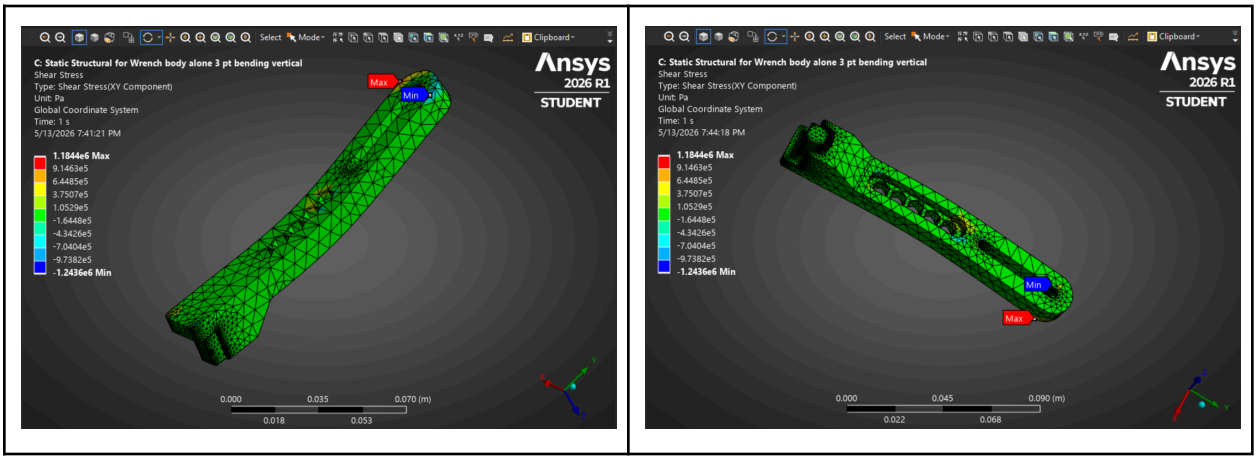


Figure X: Ansys shear stress with 3 point bending test in the z direction

3. Normal Stress

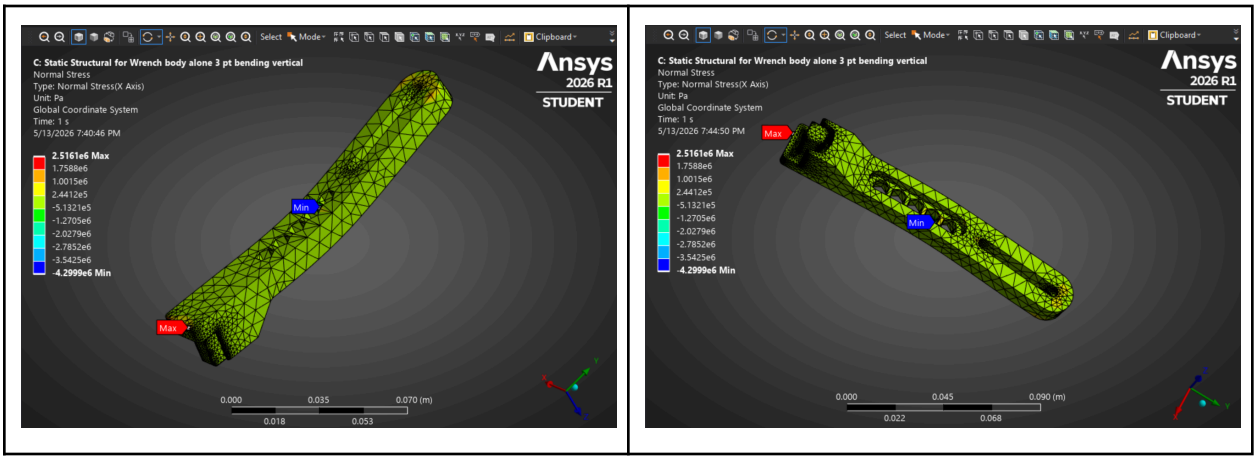


Figure X: Ansys normal stress with 3 point bending test in the z direction

4. Equivalent Stress

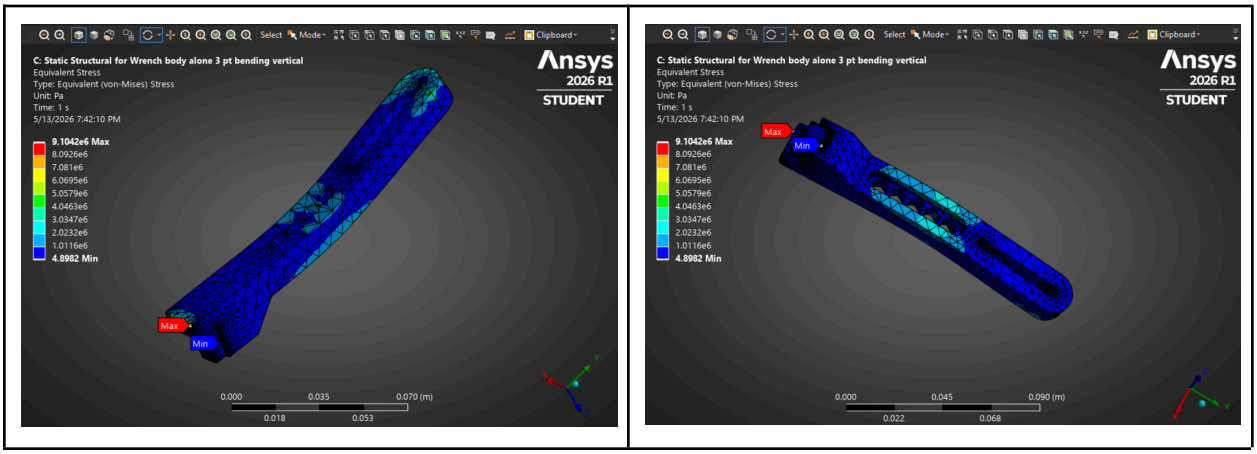


Figure X: Ansys equivalent stress with 3 point bending test in the z direction

Wrench Handle 3 point bending in x direction:

1. Total Deformation

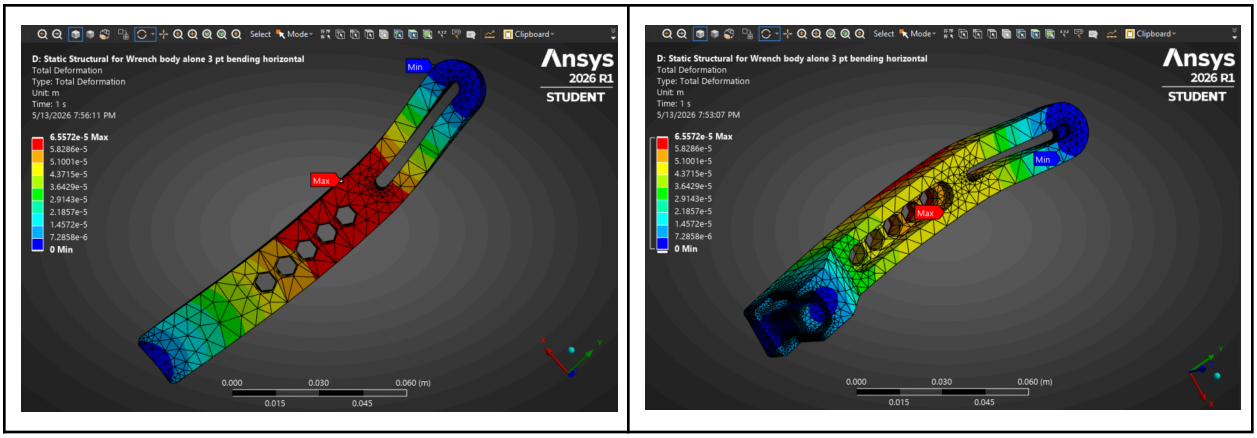


Figure X: Ansys total deformation with 3 point bending test in the x direction

2. Shear Stress

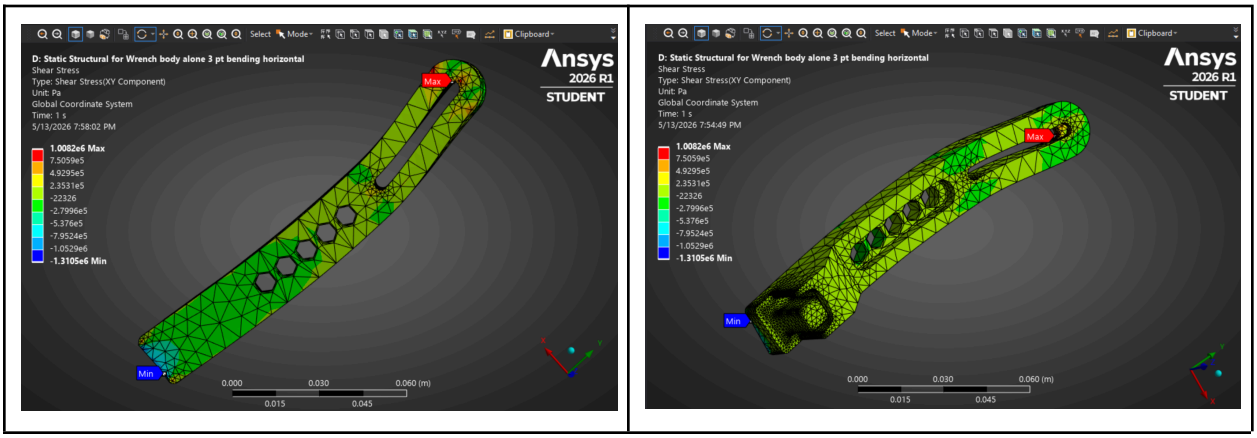


Figure X: Ansys shear stress with 3 point bending test in the x direction

3. Normal Stress

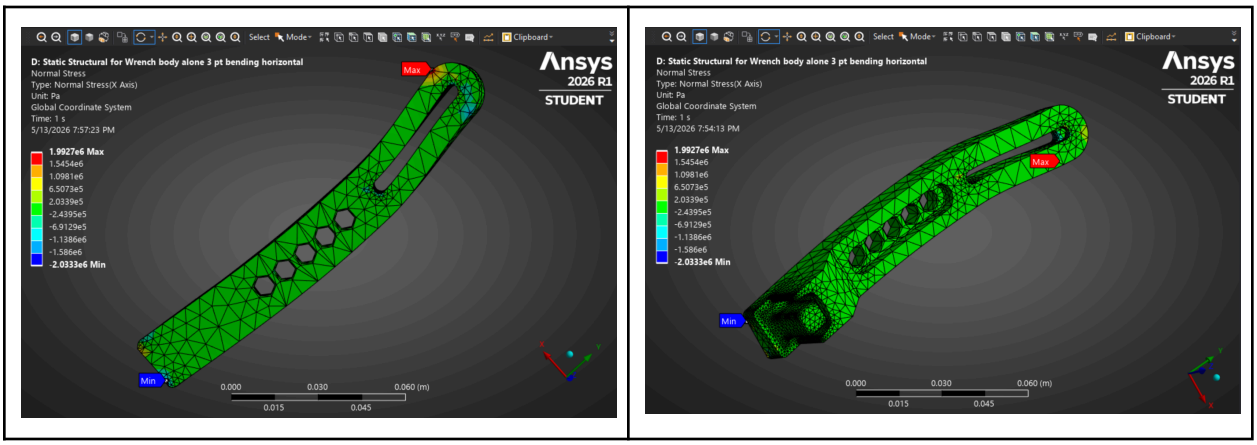


Figure X: Ansys normal stress with 3 point bending test in the x direction

4. Equivalent Stress

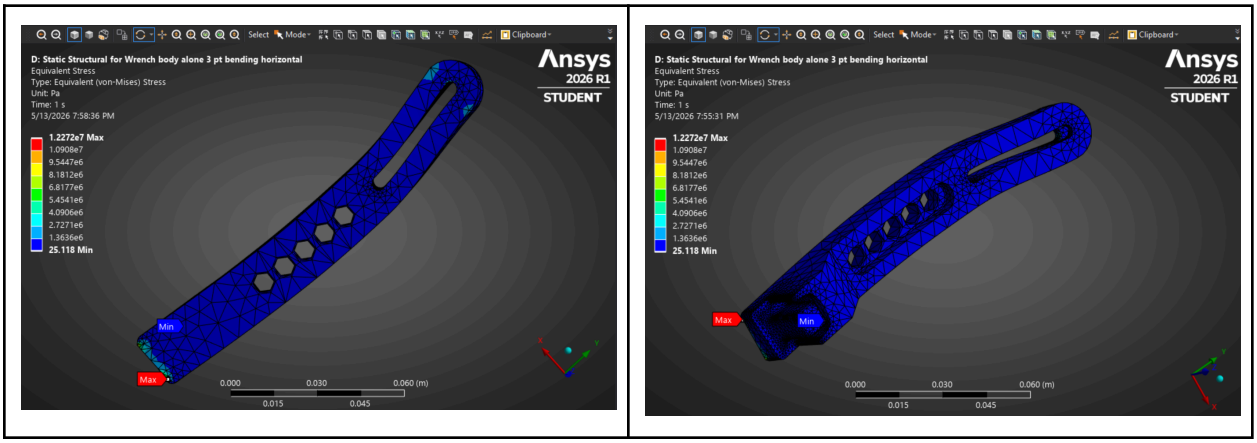


Figure X: Ansys equivalent stress with 3 point bending test in the x direction

5. Design Process:

Initial Analysis Results:

The first prototype successfully demonstrated the general concept of a lightweight multifunction wrench handle and validated the overall feasibility of the design. This initial design was printed on a 3D printer at the All Campus Makerspace at UMass out of PLA to serve as an preliminary test for fitting and identifying the weak parts in the design. The press-fit wrench head was able to insert into the handle; however, the retention was loose due to insufficient dimensional tolerances, demonstrating the importance of tighter dimensional control at the interface. During a simple manual torque and bending test, noticeable flexing occurred near the wrench head interface because of the large material removal slots. Although the handle did not fracture, the excessive deformation was larger than the group determined to be acceptable. The large cutouts reduced weight effectively but also created stress concentration regions that weakened the handle under loading conditions. Despite these structural weaknesses, the internal cavity successfully stored a screwdriver bit, confirming the feasibility of incorporating the initial secondary functionality into the design. Based on these observations, the next iteration focused on increasing structural thickness near the wrench head interface, reducing stress concentrations, and improving retention of the wrench head.

Version 2

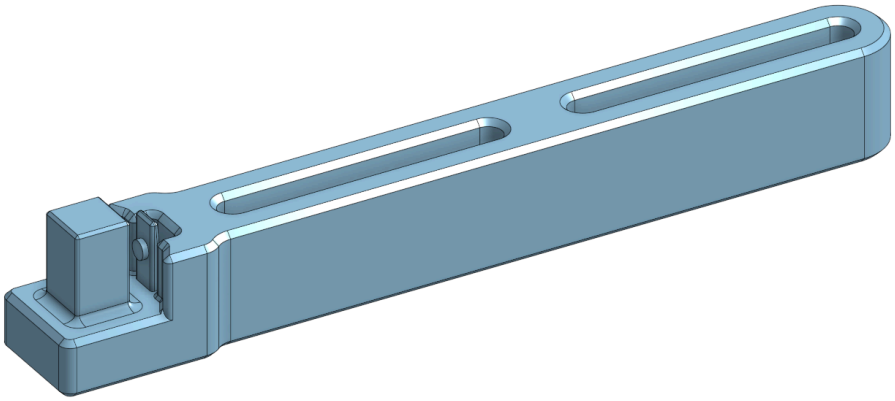


Figure X: Version 2 of Group 33 wrench handle

The second design iteration built directly upon the initial CAD model by introducing several targeted modifications intended to improve structural rigidity, wrench head retention, and overall functionality while maintaining a relatively lightweight design. One of the most significant additions was the implementation of a cantilever snap-fit feature with a circular retention pin designed to engage with the small hole located on the wrench head. This addition improved the positioning and retention of the wrench head within the handle and helped prevent unwanted movement during loading. The updated design also introduced a thicker handle body to reduce the excessive deformation observed in the first prototype and increase the overall strength of the structure under torsional loading conditions.

Additional refinements were made throughout the geometry to improve durability and reduce stress concentrations. The reduced slot size decreased the amount of material removed from the center of the handle, since the original design was already significantly below the maximum allowable mass limit. This change helped improve stiffness while still preserving the internal cavity used for screwdriver bit storage. Increased use of fillets and chamfers throughout the design further reduced localized stress concentration regions and improved load transfer between features. The revised wrench head interface also provided a secondary surface for the wrench head to react against, allowing a portion of the applied load to transfer through the surrounding wall structure rather than concentrating entirely at the original press-fit interface.

Version 3

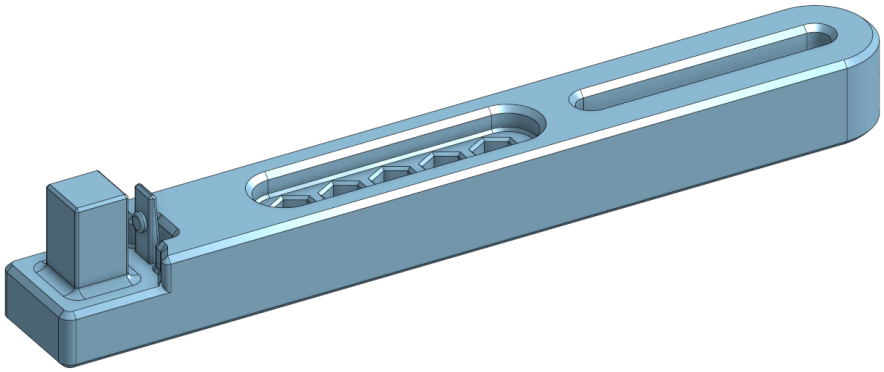


Figure X: Version 3 of Group 33 wrench handle

The third design iteration focused on dimensional precision and multifunctionality.

- Added individual holes for screw driver bits, locking their location better compared to the two wall set up
- Improved measurements of wrench head and screw driver bits for better tolerance
- Expanded upon second functionality with adding a slot at the far end of the handle such that a bit could be inserted and the wrench handle could act as a screw
- Added location for screwdriver bits to be inserted on handle and perform as a screwdriver
- Reduced thickness of handle as the modifications in version two above provided a sufficient amount of increased strength
    - Preliminary Ansys simulations showed this information
    - Weight reduction was then prioritized where material was not needed

Version 4

The fourth iteration reinforced the wrench head interface by increasing wall thickness and modifying the internal geometry.

- Thicken handle to increase surface area connection between wrench head and handle wall
- New handle wall shape dimensions to create a close fit for the wrench head
- Location of screwdriver bit holes moved to maintain the setup in the middle of the wrench
- First test print to be done on the Stratasys printers at the machine shop
    - Printed in two orientations to test the wrench strength in different directions

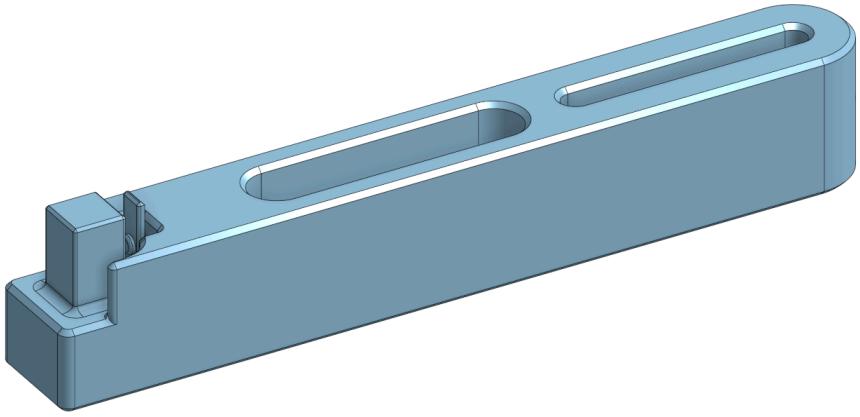


Figure X: Version 4 of Group 33 wrench handle

Final Design

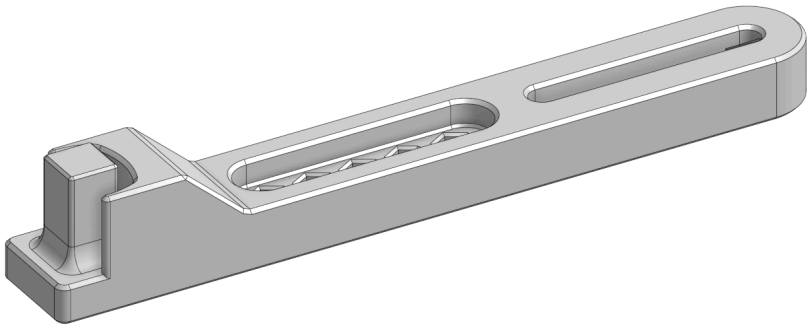


Figure X: Final version of Group 33 wrench handle

The final design iteration replaced the cantilever retention system with a simplified press-fit interface

- Removed cantilever for holding wrench head position, switched to press fit
    - Found cantilever was ineffective at deforming the proper amount without breaking

    - A tight fit from the preexisting walls already lead to a snug fit for both of the wrench heads
    - Cantilever resulted in more complexity printing and less strength
- Thickness of wrench reduced where not needed
    - Wrench handle was reduced due to ansys simulations not exceeding a FOS for the gripping part of the wrench
    - Thickness around the area that came in contact with the wrench head was not changed to provide for an acceptable press fit and for there to be more surface area for force to travel between the wrench head and the handle
    - Chamfer was implemented to ease the transition between the two thicknesses
- Screwdriver bit holes now flush with bottom of wrench to aid in 3D printing
    - Makes 3D printing easier
    - Less supports needed, decreasing post processing time
- Small changes in dimensions to allow for better fit of wrench head and screwdriver bits
- Simulated CAD weight: 33.85264 g
- Total of 6 locations for screwdriver bits
    - Tight positioning for a firm fit - cannot drop out by accident
- Ansys simulation data:
    - Bolt torque
        - Max stress:
        - Max deformation:
    - Bolt torque + applied load of 25N
        - Max stress:
        - Max deformation:
    - 3 point bending test in the z direction
        - Max stress:
        - Max deformation:
    - 3 point bending test in the x direction
        - Max stress:
        - Max deformation:
- For the torque tests, the max stress always appeared at the base of the pin.
    - Very difficult to change after several iterations and a fillet size that was extremely large (r = 5mm)
- Majority of the rest of the handle was typically always below 10MPa, with the increase in stress focused on the fillet only

6. Test Results:

Multiple design iterations were manufactured and evaluated throughout the development of the multifunction wrench handle. Testing focused on structural performance, wrench head retention, manufacturability, weight reduction, and the effectiveness of the secondary screwdriver functionality. Each iteration improved upon issues identified in the previous design and contributed to the development of the final prototype.

7. Discussion and Conclusion:

The iterative development process played a critical role in achieving a successful final design. Each prototype identified limitations and opportunities for improvement, allowing the team to refine the geometry, retention system, and structural layout of the multifunction wrench handle.

The V2 design demonstrated that increasing wall thickness and reducing excessive material removal improved stiffness and torsional resistance. However, the cantilever snap-fit mechanism introduced high stress concentration regions that reduced long-term durability. These observations highlighted the importance of minimizing localized stresses in FDM-manufactured ABS components.

The V3 design improved dimensional tolerances and introduced dedicated screwdriver bit storage and screwdriver functionality. This iteration demonstrated the value of precise interface geometry for reliable press-fit retention and effective torque transfer. The improved organization of screwdriver bits also enhanced the multifunctionality of the design.

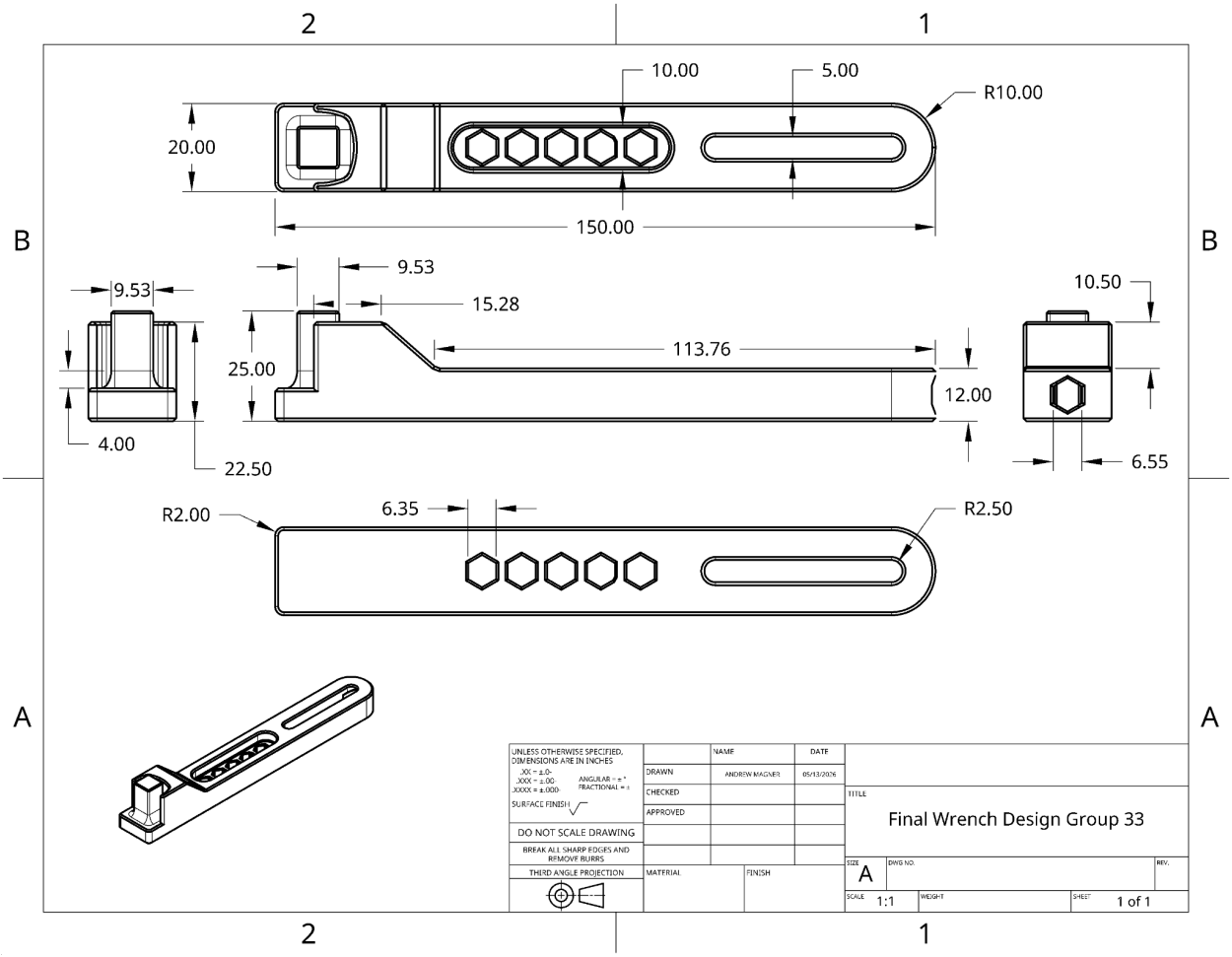
The V4 design focused on reinforcing the wrench head interface region after FEA and physical testing showed that this area experienced the highest stresses during operation. Increasing the wall thickness near the interface significantly reduced elastic deformation and improved load transfer between the handle and wrench head.

The final V5 design successfully combined the strengths of the previous iterations while simplifying the retention system. Replacing the cantilever mechanism with a press-fit design eliminated unnecessary stress concentration regions and improved manufacturability. Material was strategically removed from low-stress areas based on by-hand calculations and finite element analysis, allowing the design to remain lightweight while maintaining structural integrity.

8. References

[1] Stratasys, *"ABS-M30 Material Data Sheet,"* September 2021. [Online]. Available: https://www.stratasys.com/siteassets/materials/materials-catalog/fdm-materials/abs-m30/mds_fdm_abs-m30_0921a.pdf. [Accessed: May 13, 2026]. (stratasys.com)

Appendix: Detailed Engineering Drawings of final product



Appendix: Manufacturing Plan for final product

The final component is a custom-designed multifunction wrench handle manufactured using Fused Deposition Modeling (FDM) additive manufacturing on either a Stratasys F170 or Stratasys F190 system. The component is printed using ABS thermoplastic filament with 100% infill density to maximize structural stiffness and strength under torsional and bending loads encountered during wrench operation. The design incorporates removable support structures printed from a secondary soluble support material to enable fabrication of internal features and overhangs without compromising print quality.

The wrench handle is modeled in CAD software and exported as a .sldprt file. This file is then delivered to the machine shop operators who confirm the design is suitable for printing. It is then the machine shop operators who convert the .sltprt file to a file type that is best suited for the Stratasys software. This sliced model is then sent to the 3D printers for manufacturing.

Print Parameters:

Printer	Stratasys F170 or F190
Build Process	FDM (Fused Deposition Modeling)
Model material	ABS (Acrylonitrile Butadiene Styrene)
Infill	100% Solid
Layer Height	Standard/default machine setting
Support material	SR-30 soluble support material
Build Orientation	Optimized for torsional loading and minimizing support usage

Following fabrication, the wrench handle underwent a post-processing procedure to remove the sacrificial support structures generated during the FDM printing process. Because the component was printed using ABS on the Stratasys F170/F190 platform, the supports were manufactured using SR-30 soluble support material, which is designed specifically for chemical dissolution. The printed components were submerged in a heated aqueous support removal bath where the SR-30 material was dissolved without damaging the ABS model material. After removal from the bath, the parts were rinsed with clean water and dried thoroughly before inspection. A visual inspection was then performed to identify any print defects such as warping, incomplete support removal, surface irregularities, or layer separation that could negatively impact functionality or structural integrity.

Following post-processing, the parts underwent dimensional and functional verification testing. Critical dimensions were inspected using calipers to ensure that the interfaces for the 7/16 in and 1/2 in crowfoot wrench heads matched the intended geometry and provided adequate

engagement. The wrench heads were inserted into the handle and tested for fit, retention, and ease of installation/removal to verify proper load transfer capability during use. The integrated secondary functions were also evaluated, including the five screwdriver bit storage holes and the screwdriver slot incorporated into the base of the handle. The storage holes were tested for proper retention and accessibility of standard screwdriver bits, while the screwdriver slot was tested to confirm proper bit engagement and manual torque transfer capability.